

TOPEKA BILLARD, KS, USA

WMO: 724560

Lat: 39.0722N

Lon: 95.6306W

Elev: 269

StdP: 98.14

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13996

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-12.9	-20.9	0.6	-14.1	-18.5	0.8	-11.8	11.3	1.1	10.3	1.2	3.4	310	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	36.8	24.6	35.0	24.5	33.3	24.0	26.3	33.2	25.6	32.5	25.0	31.5	4.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.9	30.1	23.6	19.0	29.4	22.9	18.3	28.8	83.0	33.3	80.1	32.6	77.2	31.5	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.9	8.0	-19.7	39.1	2.7	1.9	-21.7	40.4	-23.2	41.5	-24.8	42.6	-26.7	43.9
			-20.2	27.6	2.6	0.8	-22.0	28.2	-23.6	28.7	-25.0	29.1	-26.9	29.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	-0.9	1.5	7.5	13.2	18.8	24.2	26.6	25.6	20.9	14.1	7.1	1.0	(6)
	DBStd	10.87	6.29	6.52	6.37	5.24	4.56	3.47	3.32	3.39	4.49	5.18	5.44	5.79	(7)
	HDD10.0	1156	341	245	123	27	1	0	0	0	1	19	117	283	(8)
	HDD18.3	2679	597	471	339	168	52	2	0	1	24	151	338	538	(9)
	CDD10.0	2386	2	8	45	122	275	427	514	484	328	147	30	4	(10)
	CDD18.3	865	0	0	3	13	67	179	256	226	101	19	1	0	(11)
	CDH23.3	9313	0	1	33	154	636	1863	2985	2439	1006	187	8	0	(12)
	CDH26.7	4164	0	0	4	33	203	791	1498	1190	393	50	1	0	(13)
(14) Wind	WSAvg	3.4	3.5	3.8	4.2	4.4	3.6	3.4	2.8	2.6	2.9	3.3	3.5	3.4	(14)
(15) Precipitation	PrecAvg	899	23	30	60	87	117	128	96	104	93	76	46	37	(15)
	PrecMax	1547	68	89	214	221	300	386	279	284	323	184	159	109	(16)
	PrecMin	485	0	1	3	16	10	14	13	7	15	1	0	1	(17)
	PrecStd	207	16	22	42	48	59	74	70	65	66	48	40	28	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	22.3	26.7	30.0	33.1	36.3	39.2	39.4	35.8	30.8	24.5	19.2	(19)
		MCWB	10.7	12.4	16.9	18.2	21.9	24.5	24.2	24.2	23.3	21.1	15.9	14.1	(20)
	2%	DB	14.6	18.3	23.6	27.1	30.9	34.1	37.0	36.5	32.9	27.9	21.5	15.4	(21)
		MCWB	8.8	11.2	14.8	17.4	21.5	24.4	24.9	24.3	22.9	18.8	14.6	10.4	(22)
	5%	DB	11.4	15.1	20.9	25.0	29.1	32.6	35.3	34.6	31.0	25.2	18.8	12.6	(23)
		MCWB	6.8	9.5	13.8	16.4	20.6	23.9	25.1	24.2	22.4	17.9	12.4	8.5	(24)
	10%	DB	8.5	11.9	18.1	22.6	27.0	31.1	33.4	32.7	28.8	22.9	16.2	9.9	(25)
		MCWB	4.5	7.3	12.1	15.7	19.9	23.1	24.6	23.8	21.1	17.0	11.5	6.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	15.1	18.8	20.6	24.1	26.7	27.4	26.8	25.2	22.7	17.6	15.5	(27)
		MCDB	15.6	18.6	24.5	26.7	30.7	33.1	34.2	33.6	31.7	28.0	22.3	18.3	(28)
	2%	WB	9.1	12.2	16.8	19.3	23.0	25.6	26.5	25.9	24.1	21.0	15.6	11.8	(29)
		MCDB	13.6	16.5	21.2	24.8	28.7	32.1	33.4	33.0	30.3	25.4	19.7	14.2	(30)
	5%	WB	6.9	9.8	14.9	17.8	21.8	24.7	25.7	25.2	23.3	19.4	14.0	8.6	(31)
		MCDB	11.2	15.4	19.5	23.4	27.2	31.0	32.7	32.0	29.1	23.4	17.5	12.3	(32)
	10%	WB	4.7	7.1	12.6	16.2	20.7	23.9	25.2	24.5	22.3	17.7	11.9	6.4	(33)
		MCDB	8.4	11.7	17.8	21.7	25.7	29.8	32.0	30.7	27.7	21.7	15.7	10.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.2	12.2	12.2	11.3	10.9	11.1	11.6	12.3	12.4	11.8	10.3	(35)
		MCDBR	16.3	16.9	16.7	15.6	13.4	12.0	12.7	13.7	13.5	14.9	15.8	15.1	(36)
		MCWBR	11.0	10.9	9.6	8.5	6.2	4.9	4.3	4.7	5.5	7.6	9.5	10.4	(37)
	5% WB	MCDBR	15.4	15.6	13.1	13.3	11.4	11.1	11.4	11.9	11.6	11.5	12.9	13.5	(38)
		MCWBR	10.9	10.8	8.4	8.3	6.2	5.3	4.7	4.9	5.4	7.0	9.1	10.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.289	0.303	0.333	0.386	0.420	0.432	0.435	0.435	0.387	0.342	0.306	0.285	(40)
	taud		2.479	2.450	2.422	2.294	2.261	2.257	2.274	2.278	2.362	2.474	2.514	2.508	(41)
	Ebn at Noon		893	925	926	890	863	850	843	834	856	867	865	872	(42)
	Edh at Noon		78	92	106	128	135	136	132	129	111	90	75	71	(43)
(44) All-Sky Solar Radiation	RadAvg		2.26	3.06	3.96	5.03	5.77	6.53	6.48	5.78	4.80	3.48	2.48	1.94	(44)
	RadStd		0.22	0.33	0.32	0.33	0.50	0.43	0.44	0.29	0.33	0.39	0.22	0.18	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page